Vision-Guided Terrain-Aware Path Planning for Autonomous Rovers

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Abstract

It is essential for autonomous ground robots operating in unstructured environments to perform reasoning not only for obstacle avoidance but also for the assessment of the physical difficulty of the terrain they move on. The majority of the existing motion planning pipelines are based on binary traversability maps or static cost representations which cannot reflect the continuous variations in the terrain properties like slope, surface friction, and roughness. This limitation frequently results in suboptimal path selection, higher energy consumption, and unstable vehicles. We, therefore, propose a framework for path planning that is both terrain-aware and vision-guided for rover-type autonomous vehicles. This integrates semantic perception with cost modeling based on physics. A convolutional neural network takes camera input from the onboard system and does real-time semantic segmentation to determine terrain types. The semantic predictions are then merged with the geometrics of the terrain and the properties of the materials that have been acquired from an analog simulation environment in Gazebo, this enables the creation of a continuous, physics-aware terrain difficulty cost map. The resulting cost map shows risk of traversal and energy expending more accurately than traditional binary representations. A graph-based planner that uses A* and RRT* is implemented and that's how minimum cost paths are computed dynamically with the help of difficulty estimating and terrain based distance rather than just geometric distance being taken into account. The system we propose is tested in off-road situations that have different kinds of terrain with slopes and low-friction areas. The results of the experiments indicate that planning with awareness of the terrain greatly decreases the cost of the whole traversal and increases the safety of the path selected as compared to the planners running on either uniform or obstacle-only cost maps.

Problem Statement

Autonomous navigation for extraterrestrial rovers remains a fundamentally challenging problem due to the interaction between perception uncertainty, terrain physics, energy constraints and reduced gravity environments. Current rover navigation pipelines predominantly rely on two-dimensional (2D) representations of the environment, such as occupancy grids derived from LiDAR-based SLAM. While computationally efficient, these representations implicitly assume planar traversability and binary obstacle semantics, which fail to capture the continuous nature of terrain difficulty encountered on planetary surfaces.

Mathematical Limitations of Current Models

Formally, most existing navigation systems model the environment as a binary grid map:

$$M_{2D} : \mathbb{R}^2 \rightarrow \{0, 1\} \tag{1}$$

where cells are classified as either free or occupied. Path planning then minimizes Euclidean or Manhattan distance:

$$\pi^* = \arg\min_{\pi} \sum_i \|x_{i+1} - x_i\| \tag{2}$$

Such formulations ignore elevation gradients, material properties, and traction dynamics. In multilevel or sloped environments, this abstraction produces physically infeasible paths, particularly when vertical discontinuities, ramps, or loose regolith are present. As a result, planners may select routes that appear optimal in 2D but violate rover mobility constraints.

Environmental and Energy Challenges

These limitations are amplified in extraterrestrial settings characterized by heterogeneous materials and varying slopes. Moreover, gravity differs significantly from Earth (e.g., $g_{\text{Moon}} \approx 0.165 g_{\text{Earth}}$, $g_{\text{Mars}} \approx 0.38 g_{\text{Earth}}$), altering normal forces and, consequently, traction. Our simulation setting tries to bridge the gap by testing and adjusting it for any such unseen circumstances.

Energy efficiency further complicates the problem. Existing planners typically optimize for geometric shortest paths rather than minimizing cumulative energy expenditure, where:

$$E(\pi) \neq \sum_i \|x_{i+1} - x_i\| \tag{3}$$

Instead, $E(\pi)$ depends on terrain-dependent resistance, slope-induced torque, and slip losses.

The Research Gap

Therefore, the core problem addressed in this work is the absence of a unified, physics-aware navigation framework that:

- Models terrain traversability as a **continuous cost function** rather than a binary constraint.
- Incorporates **semantic perception**, slope, and material-dependent traction.
- Explicitly accounts for **energy consumption** and battery conservation.
- Remains **adaptable** across varying gravitational environments.

Addressing this gap is essential for enabling robust, energy efficient, and autonomous navigation of extraterrestrial rovers in complex and uncertain terrain.

Notation and Formula Definitions

The following notations define the models and objectives used in this framework:

M_{2D} **2D Occupancy Grid:** A mapping function $M_{2D} : \mathbb{R}^2 \to \{0, 1\}$ that represents the environment as a discrete grid.

$\{0, 1\}$ **Binary States:** The classification of cells where 0 represents free space and 1 represents an occupied obstacle.

π^* **Optimal Path:** The sequence of waypoints that minimizes the cost function, often expressed as $\arg\min$.

$\|x_{i+1} - x_i\|$ **Euclidean Norm:** The geometric distance calculated between two consecutive points x_i and x_{i+1}.

$E(\pi)$ **Cumulative Energy:** The total energy cost, which is functionally different from distance ($\neq \sum \|\Delta x\|$).

$g_{\textbf{Moon/Mars}}$ **Gravitational Constants:** The acceleration due to gravity on the target body ($g \approx 0.165 g_E$ for Moon), impacting traction.

Background & Literature Review

The rapid expansion of autonomous systems, encompassing both Unmanned Ground Vehicles (UGVs, or Rovers) and Unmanned Aerial Vehicles (UAVs or Drones), into complex, unstructured, and often hazardous environments such as forests, disaster relief zones, or planetary exploration sites necessitates a fundamental re-evaluation of motion planning architectures. Traditional methods, largely optimized for distance minimization and geometric collision avoidance, are insufficient when navigating terrains where success is governed by complex physical interactions. The transition to terrain-aware navigation demands a move beyond simple two-dimensional (2D) or even 2.5D planning into true three-dimensional (3D) space, where variables like slope, ground friction, and dynamic stability fundamentally determine path feasibility and risk. A primary deficiency in current architectures is the inherent trade-off observed between optimizing for path length and minimizing physical cost. Research demonstrates that simply seeking the shortest path often results in maximizing energy consumption or incurring significant stability risk. A successful terrain-aware planning system must resolve this conflict by developing multi-objective optimization strategies that seamlessly fuse geometric metrics with critical physics-based costs, ensuring both safety and mission endurance. For the proposed project involving heterogeneous robotics (Drones and Rovers), this challenge is amplified. Rovers require fine-grained ground-contact physics modeling (traversability, wheel slippage) over short distances, while Drones require volumetric mapping capabilities to determine air path feasibility, obstacle clearance, and safe landing/take-off zones over vast distances. The successful integration of perception and planning across these vastly different scales and data modalities constitutes a critical technical challenge that previous single-platform studies have not addressed.

Historically, traversability assessment relied on binary classification: areas were either deemed walkable or non-walkable based primarily on coarse geometric features such as maximum slope angle or step height. Modern research, however, emphasizes the need for a continuous risk quantification. This progression has led to two main methodological avenues: data-driven methods that learn risk from robot behavior, and physics-infused methods that attempt to estimate fundamental physical parameters of the terrain. The following review examines representative state-of-the-art works across these paradigms to identify the technical limitations that justify the proposed Vision-Guided Terrain-Aware Path Planning project.

Recent work in autonomous motion planning increasingly focuses on coupling visual perception with terrain-aware navigation to achieve safe and efficient movement across unstructured environments. (Feng et al. 2024) introduced URA*, an uncertainty-aware path planner that employs an ensemble of CNNs (U-Net and DeepLab V3+) to infer pixel-wise traversability probabilities from aerial imagery. The resulting probabilistic cost maps are integrated into an A*/D*-based search that replans under perception uncertainty. While URA* demonstrates strong perception-planning coupling, its representation of terrain remains discrete and probabilistic, lacking the continuous, physics-based cost modeling such as slope, friction, and roughness that directly relates to energy use and vehicle stability.

(Wang, Huang, and Zhao 2025) presents a real-time 3D vision-based autonomous navigation framework for wheeled robots, integrating RGB-D SLAM for mapping and localization with the A* pathfinding algorithm executed over dynamically updated NavMeshes. The system utilizes vision-derived obstacle detection to ensure robust collision avoidance and continuous path replanning in dynamic 3D environments. This work validates the feasibility of the project's architecture, particularly the Path Planning (A*) step which employs A* on a cost map derived from 3D vision. The proposed project directly addresses this paper's core limitation by generating a Dynamic Terrain Difficulty Estimation Cost Map—a physics-aware and continuous cost layer—and injecting it into this established A* planning framework, moving beyond simple geometric avoidance toward risk/energy minimization.

(NanWang and Xie 2024) A Survey on Path Planning for Autonomous Ground Vehicles (2023) emphasizes that path planning forms the backbone of autonomous navigation, enabling vehicles to generate collision-free, dynamically feasible, and efficient trajectories across diverse environments.

The study outlines how modern planners integrate perception, environment modeling, and cost-based optimization to handle varying terrains, obstacles, and motion constraints. It categorizes approaches such as graph-based (A*, D*) and sampling-based (RRT, PRM) methods, analyzing their adaptability to both structured and off-road scenarios. Importantly, the survey highlights the growing trend toward terrain-aware and energy-optimized planning frameworks that account for slope, traction, and risk. These insights strongly support our project's approach of coupling vision-based terrain difficulty estimation with A*/RRT* path planning for more robust and energy-efficient rover navigation. (Wei and Isler 2020) presents a data-driven approach for UGV path planning that stochastically models energy consumption using a combination of aerial images and ground measurements. By correlating visual inputs with energy costs across various terrain surfaces, the method enables the use of constraint-aware path planning heuristics to find energy-efficient routes. This is highly relevant as it explicitly validates the project's use of a hierarchical, air-ground sensing approach to inform energy-efficient UGV navigation. Our project takes the concept further by using the Drone's visual system (DeepLabV3+ segmentation) to predict explicit physics-aware parameters (Slope, Friction) rather than stochastic correlations, thus providing a more direct and interpretable Dynamic Terrain Difficulty Estimation input for the A*/RRT* planner.

(Yang et al. 2017) Autonomous navigation for planetary rovers remains a fundamental challenge due to the complex interaction between perception, terrain geometry, vehicle dynamics, and energy constraints. Most operational rover missions on Mars and the Moon, including NASA/JPL's *Spirit*, *Opportunity*, *Curiosity*, and *Perseverance*, have relied heavily on ground-in-the-loop planning and conservative mobility strategies to mitigate risk. Despite these precautions, multiple missions have experienced mobility-related failures, such as wheel degradation on sharp Martian rocks, excessive sinkage in soft regolith, and irreversible entrapment in sandy terrain. These failures highlight the inadequacy of navigation systems that do not explicitly model terrain-dependent traversability and physical constraints. Traditional robotic navigation pipelines typically employ two-dimensional SLAM and occupancy grid–based planning, where the environment is represented as a binary free/occupied map. While effective in structured terrestrial settings, such representations break down in multilevel or deformable environments. In reduced-gravity extraterrestrial settings, a terrain cell that appears geometrically traversable may be infeasible due to slope instability, low friction coefficients, or excessive energy expenditure. Let $\mathcal{M}(x, y)$ denote a 2D occupancy map and $\mathcal{C}(x, y)$ the corresponding traversal cost. Classical planners assume $\mathcal{C}$ depends primarily on obstacle proximity, ignoring physical attributes such as local slope $\nabla h(x, y)$, surface friction $\mu(x, y)$, and gravity g. Recent learning-based approaches attempt to address perception limitations by mapping sensor data to intermediate representations, such as depth or semantic labels, prior to planning. While these methods improve interpretability, they remain largely geometry-centric and do not incorporate physics-aware cost modeling. Moreover, they assume Earth-like gravity and surface interaction, which is invalid for lunar or Martian environments. On the Moon, for example, reduced gravity lowers normal force, which paradoxically reduces traction on loose gravel, increasing slip despite lower vehicle weight. In addition to mobility risk, energy efficiency is a critical constraint for extraterrestrial rovers operating under strict battery and thermal budgets. Let $E(\pi)$ denote the energy consumed along trajectory π. Minimizing path length alone does not minimize $E(\pi)$ when terrain-dependent resistance and slope-induced power draw dominate energy usage. Current navigation frameworks fail to jointly optimize safety, feasibility, and energy consumption. This work addresses these limitations by formulating navigation as a physics-aware optimization problem that integrates semantic terrain perception, slope estimation, gravity-adjusted traction modeling, and energy-aware cost functions. By moving beyond purely geometric maps toward continuous difficulty-aware cost representations, the proposed framework enables safer, more autonomous, and more energy-efficient navigation for planetary rover missions.

Technical gaps and challeneges

Even though vision-based navigation and cost-aware motion planning have advanced recently, there are still some gaps that limit the reliable use of these techniques in rough terrains. Firstly, a number of methods used today depend on either binary or discretized traversability maps which do not take into account the continuous variations in the difficulty of the terrain resulting from slope, surface friction and roughness—factors that directly influence the stability and energy consumption of the vehicles. Secondly, although learning-based perception methods are capable of deriving terrain semantics, the resulting predictions are often not syncing with the physics-based reasoning, thus, the cost representations lack physical interpretability and generalization across environments. Thirdly, the fusion of different sources such as visual perception, terrain geometry, and material properties into a single, real-time planning cost map is still a challenge due to the discrepancies in spatial resolution, uncertainty and update rates. Furthermore, the dynamically updating terrain-aware cost maps lead to computational and latency constraints, especially when combined with graph-based planners that need to replan efficiently in response to the changing terrain conditions. Finding solutions to these difficulties demands a planning framework that will be able to manage physical realism, computational efficiency and environmental variability.

Methodology

Methodological Overview

This project establishes a physics aware path planning framework specifically engineered for autonomous rovers (UGVs) operating in unstructured, off-road environments. The primary objective is to transcend the limitations of traditional geometric planners which view the world merely as "occupied" or "free" by integrating a "proprioceptive" understanding of the physical world. By fusing visual semantic data with simulated physics, the system optimizes navigation for energy efficiency and risk minimization, effectively teaching the rover to identify and avoid "high-effort" terrain before it commits to a path.

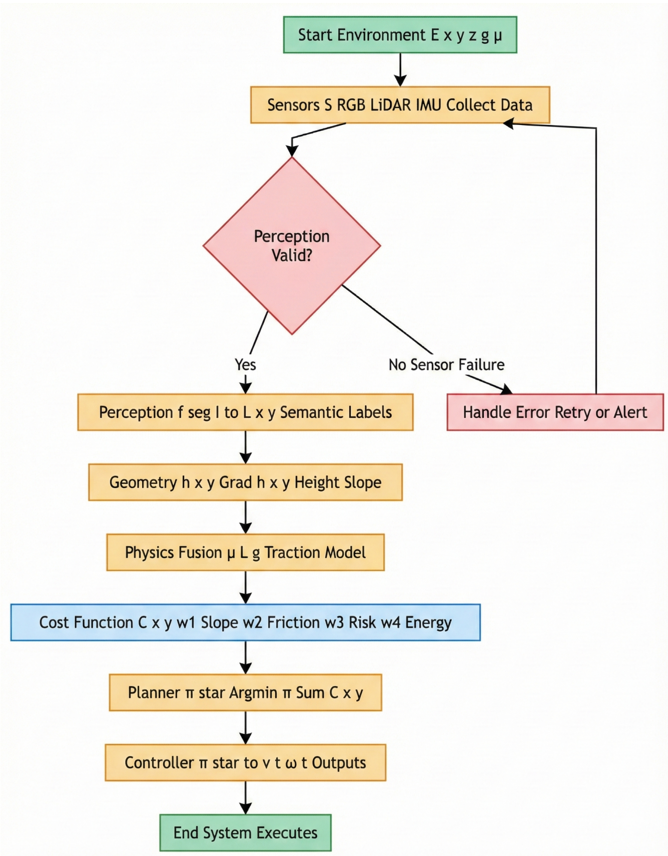


Figure 1: Methodology Flowchart

Visual Perception via Convolutional Neural Networks (CNN)

The first stage of the pipeline focuses on scene understanding. We implemented a **Convolutional Neural Network (CNN)**, specifically the **DeepLabV3+** architecture, to perform real-time semantic segmentation of the rover's visual feed. A **ResNet18** backbone was selected for this architecture to provide an optimal balance between segmentation accuracy and the computational efficiency required for onboard deployment.

To address the challenge of generating training data for unstructured environments, we developed a semi-supervised "Auto-Labeling" algorithm. This pipeline utilizes HSV color physics to automatically generate Ground Truth masks from raw Gazebo simulation imagery. This allowed the CNN to learn and classify distinct terrain types crucial for navigation, specifically identifying **Sand** (traversable), **Mud** (high friction/risk), and **Grassland** (roughness/obstacle). This semantic layer provides the necessary context for the physics engine to estimate surface friction.

Physics-Aware Dynamic Cost Map Generation

The core novelty of this framework is the **Dynamic Cost Map**, which translates the semantic labels from the CNN and geometric data into a unified quantitative difficulty score. The map utilized for this project was designed to represent **real-world terrain topology**, incorporating complex elevation changes and surface varieties that mimic actual field conditions.

The cost estimation logic uses a weighted fusion approach, assigning **70% weight to Height (Slope)** and **30% weight to Friction**. Unlike standard planners that assign an absolute cost to a slope regardless of the robot's direction, we introduced a novel **Directional Slope Cost**. This model explicitly accounts for **gravity assistance**, treating descent as "free energy." Consequently, the system reduces the traversal cost when the robot is descending, while penalizing ascent.

The specific formula derived for this cost estimation is:

$$Cost_{total} = 0.3 \cdot \mu + 0.7 \cdot |\Delta h| \cdot (1 + \lambda \cdot \mathrm{sgn}(\Delta h)) \quad (4)$$

Where μ represents the friction coefficient, Δh is the elevation change, and the term $\lambda \cdot \mathrm{sgn}(\Delta h)$ applies the directional bias.

Visualization and Verification with Matplotlib

To ensure the accuracy of the physics model and the segmentation logic, the generated cost maps, elevation profiles, and terrain classification layers were visualized and rigorously verified using **Matplotlib**. These visualizations were critical for debugging the layering of friction and slope costs, allowing us to visually confirm that the low-cost areas on the heat map corresponded correctly to safe, downhill, or flat terrain, while high-cost areas accurately reflected steep ascents or high-friction mud traps.

Hierarchical Path Planning Architecture

The navigation stack utilizes a two-tiered architecture.

- **Global Planner:** We implemented and benchmarked both **A*** (A-Star) and **RRT** (Rapidly-exploring Random Tree) algorithms to calculate the optimal path over the static cost map. The A* planner proved particularly effective at exploiting the physics-aware map, generating paths that "curve" around slopes to utilize gravity rather than driving linearly over high-energy ridges.
- **Local Planner:** For real-time execution, we employed the **Dynamic Window Approach (DWA)** refined with a local A* search. This layer handles dynamic obstacles and trajectory adjustments, ensuring the robot remains safe while adhering to the energy-efficient global plan.

Implementation and Results

The complete system was architected in **ROS 2** and validated in the **Gazebo** simulation environment. Performance metrics indicated that the A* + DWA configuration achieved a path length of 779.20 cells with an optimized total cost of 254.75, successfully demonstrating the system's ability to trade geometric distance for reduced physical effort and energy consumption.

Results and Discussions

Terrain Map Generation

Height Map Representation: The height map visualizes the elevation profile of the simulated environment, serving as the primary geometric representation of the terrain. Each cell in the map encodes the absolute elevation value extracted from the simulation environment, with color gradients indicating relative height differences. Regions with higher elevation and steeper gradients correspond to slopes, ridges, and uneven terrain, while smoother color transitions indicate flatter, more traversable areas. This representation captures fine-grained terrain undulations that are not explicitly visible in raw RGB imagery but are critical for assessing rollover risk and traversal difficulty. By preserving continuous elevation variations rather than discretizing the terrain into flat regions, the height map provides the necessary geometric foundation for computing slope and incline metrics used in downstream cost estimation. As shown in the figure, the elevation data reveals complex terrain structure, including gradual inclines and localized depressions, which would be inadequately represented using binary obstacle maps.

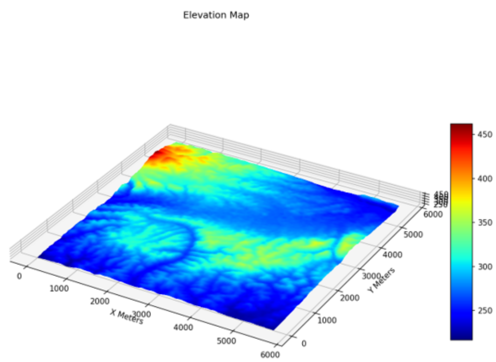


Figure 2: Height Map

Terrain Classification Map The terrain classification map represents the outcome of the semantic segmentation module where each pixel gets a class of terrains according to its visual aspect. The image portrays the model's capability of producing reasonably large areas with the same class, hence, efficiently filtering out the pixel noise that was generated in the ground-truth labels. Furthermore, the classifier is able to smoothly classify the terrains even when textures are continuous, hence creating smooth semantic boundaries even in the regions with slight visual differences. This semantic layer acts as an extra layer of information that is not revealed through the geometry alone, such as the surface's composition and texture which are the factors that directly affect the traction and energy consumption. The final terrain classification map is a very high-level perceptual abstraction that links the raw visual input with the physics-aware reasoning, thus allowing the system to make a connection between the visually-similar regions and the consistent traversal characteristics.

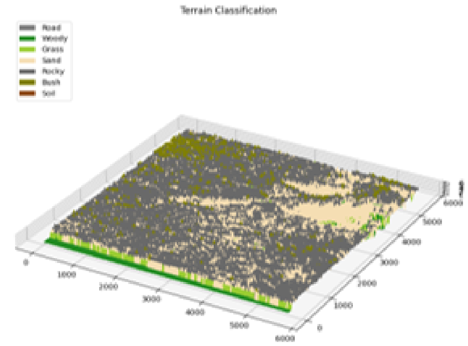


Figure 3: Terrain Classification Map

Physics-Aware Cost Map The cost map illustrates the ultimate integrated output of the recommended method that combines semantic terrain classification with geometric and physical terrain attributes. The continuous traversal difficulty value is encoded in each cell of the cost map, with higher costs considering regions with increased slope, lower friction, or higher obstacle density. In comparison to binary traversability maps, the cost map presents smooth spatial transitions, which mirror gradual changes in terrain difficulty instead of sudden feasibility boundaries. Therefore, the planner is in a position to make a reasoned decision regarding whether to choose a longer and safer path or a shorter and riskier one. As shown in the visualization, the areas of high-cost overlap with the steep slopes and low-traction surfaces marked in the height and terrain classification maps, while the pathways of low-cost unroll over the flat and stable terrain. Hence, the cost map not only reflects an interpretable and physically meaningful representation but also leads the planner to safer and less energy-consuming paths.

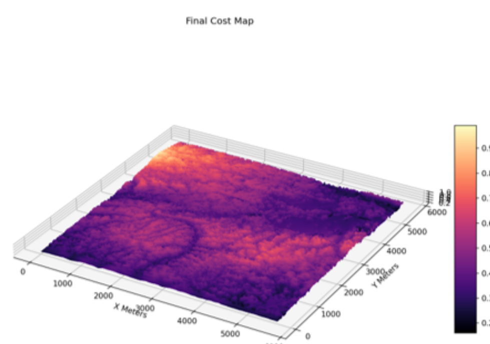


Figure 4: Physics-aware Cost Map

Relationship Between the Maps The combination of these three maps—height map, terrain classification map, and cost map—shows the step-by-step abstraction used by the suggested system. The height map shows the basic geometric structure, the terrain classification map indicates the surface properties with meaning, and the cost map combines all these modalities to give a single representation of the difficulty of traversing the area. With such a layered representation, the planning module can deal with the terrain continuously and in a physics-informed manner, which is not possible with geometry-only or semantic-only representations.

Semantic Segmentation Performance

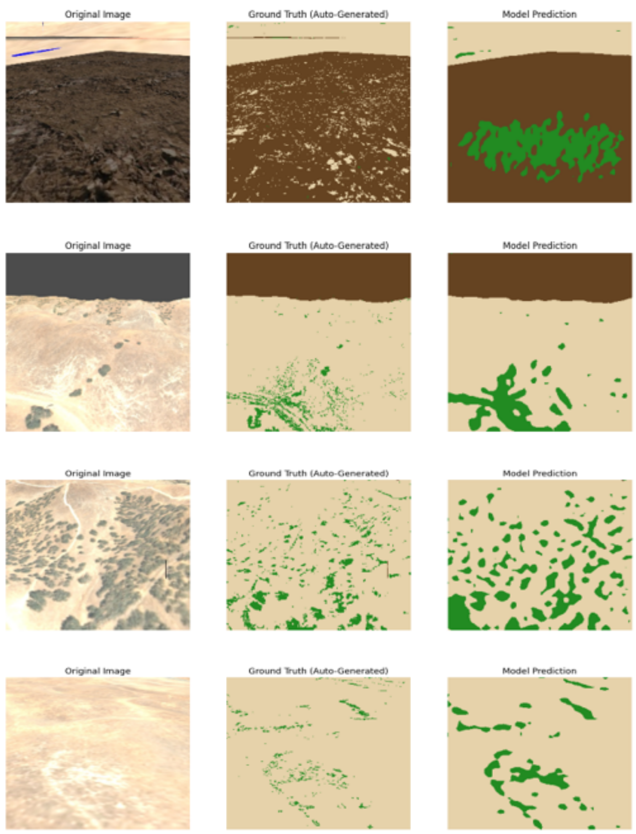


Figure 5: Comprehensive comparison of different model outputs

The qualitative segmentation outcomes pointed out a very good generalization over different types of terrain textures, which is illustrated in Fig. Y. Even though there were some pixel-level noises in the automatically generated ground-truth labels, the model still managed to learn the terrain regions coherently, especially for the continuous surfaces of mud and sand. The predictions are found to be smoother spatially and have less speckle noise in comparison to the raw labels, which is a sign of effective texture-level generalization. Moreover, the adaptive sensitivity tuning enables the model to differentiate very accurately between dry or faded vegetation and sandy terrain, which in turn leads to a significant decrease in false positives in low-contrast areas. These findings provide evidence that the perception module delivers a dependable semantic layer, which is a prerequisite for the subsequent physics-based traversability estimation.

Path Planning Execution

In Figure Z, different planning methods are compared by showing the planned paths on the same cost map with a fixed start and goal configuration. The A* planner gives a straight path that has the least cost by considering the cost already incurred while the RRT-based planner is exploring the state space with a longer and more expensive trajectory. The addition of a local refinement stage using Dynamic Window Approach (A* + DWA) results in a very small global path change while keeping cost efficiency and improving local smoothness. Remarkably, all planners have managed to completely dodge the regions with high-cost terrains which is a clear indication that the cost map not only helps the planning process to steer clear of the areas with high costs or high-energy consumption but also aids in overcoming obstacles.

Simulation Based Evaluation

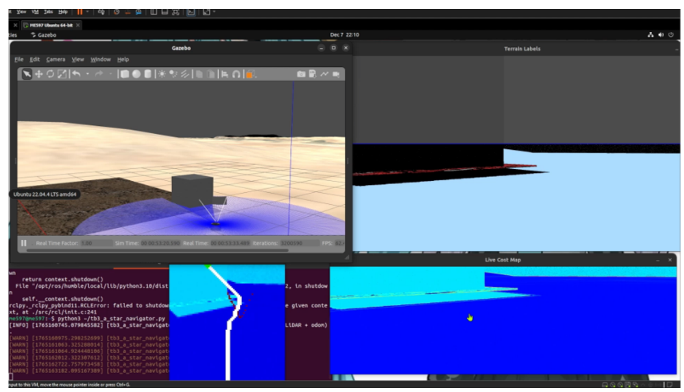


Figure 7: Gazebo Simulation

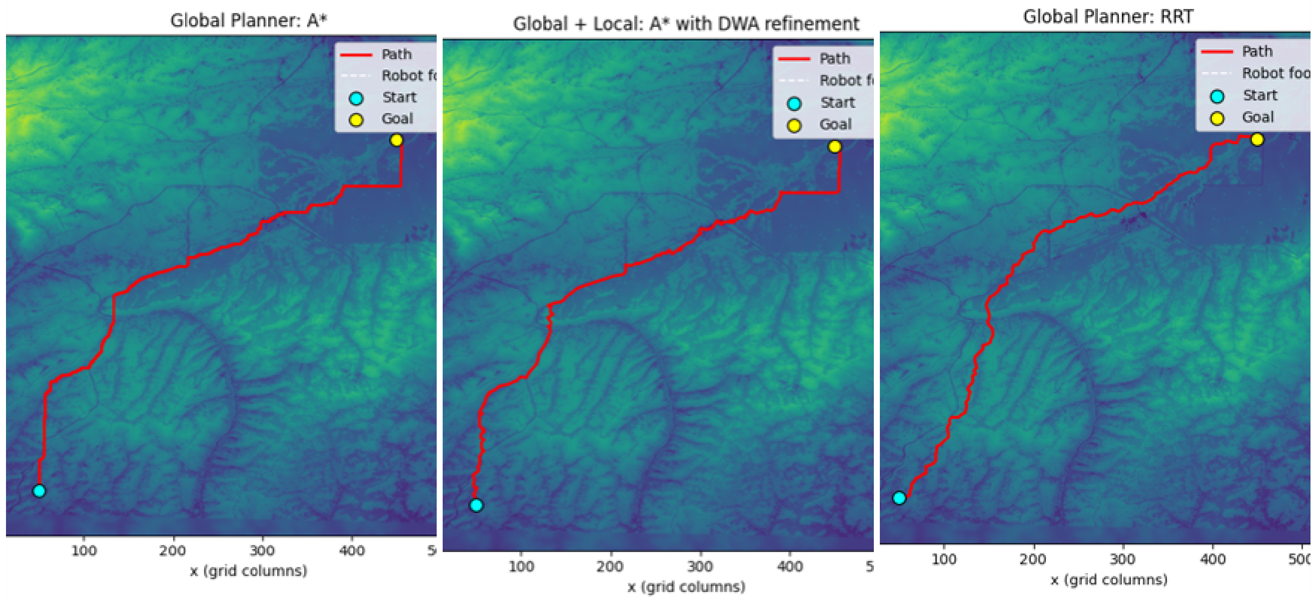
End-to-end simulation results confirm the practical viability of the proposed framework. The rover navigates the virtual world precisely along the pre-determined routes even when the difficulty of the terrain is altered. While the cost map is being live-updated, there are no sudden changes in the planning and the system behaves in a very stable manner. This shows that perception combined with cost estimation and planning is working tolerance of real-time simulation navigation.

Quantitative Analysis

Table 1: Comparison of Path Planning Algorithms

Planner	Path Length (Cells)	Total Cost	Avg. Cost	Compute Time (s)
A*	812.00	253.86	0.312	0.000178
RRT	785.18	290.46	0.328	0.000304
A* + DWA	779.20	254.75	0.314	0.000182

Table 1 presents a performance comparison of the different planners through quantitative metrics. RRT planner,



(a) A* (b) A* + DWA Refinement (c) RRT

Figure 6: Comparison of three path planning Algorithms in same conditions

though achieving a slightly shorter path length measured in grid cells, without a doubt resulted in the highest total and average traversal cost which indicates that the shorter geometric paths do not always represent the safest and most efficient routes. The opposite situation is seen with the A* planner which yields the lowest total traversal cost thus indicating its very effective manipulation of the continuous cost map. The hybrid A* + DWA strategy is bringing a good compromise by keeping the cost that is close to optimal and being able to put more emphasis on making the path smoother. The computation times of all planners are small and thus support the claim that the proposed terrain-aware cost mapping does not entail an unmanageable computational overhead.

Discussion

In summary, the findings provide validation for the main hypothesis of this research: the inclusion of physics-aware, dynamic terrain difficulty in the planning system results in safer and more economical navigation compared to the use of geometry-based or simple binary representations. Both the qualitative and quantitative assessments demonstrate that the cost map directed planners prioritize ground that reduces both risk and effort of passing through rather than merely shortening the way. The simulation was conducted in the experiment, but the application of a highly accurate physics environment guarantees that the learned and deduced terrain characteristics are still physical meaning, paving the way for the approach to be transferred to real-world robotic platforms.

Conclusion

The authors put forth a guided by vision, terrain-aware path planning method, which combined semantic perception with physics-informed cost modeling to allow navigation through carnivorous environments with high durability. The new method creates a continuous difficulty cost map of the terrain that reflects more closely the risk of traversal and energy expenditure than the traditional binary representations by merging terrain classification from visual input with geometric and material characteristics obtained from a high-fidelity simulation environment.

Experiments carried out in simulation show that planners utilizing the suggested cost map consistently stay away from the regions of high-risk terrain and, as a result, have lower total traversal costs in comparison with geometry-only and distance-based planning methods. All the qualitative and quantitative evaluations confirm that the framework can accommodate several planning techniques, such as A*, RRT, and hybrid global–local ones, without incurring significant computation time.

In conclusion, the findings signal the necessity of closely linking perception, physics-aware terrain modeling, and cost-based planning for ground autonomous navigation. Although the present implementation is assessed in simulation, the modular design and the dependence on physically meaningful terrain parameters clearly indicate a route for transfer to real-world robotic platforms. Next, the uncertainty-aware risk estimation, dynamic terrain adaptation, and real-world validation will be incorporated in the future research to further enhance the system's robustness and generalization.

Future Works

Many different ways can be pursued to amplify and fortify the proposed terrain-aware planning framework. First to mention, the use of uncertainty-aware terrain modeling would give the system the opportunity to explicitly take into account the mistakes of the perception and the unclear terrain regions. Degrading the visual environment's robustness and pretending the environment is only partially perceptible may be achieved through using probabilistic cost representations that include risk-sensitive metrics or Conditional Value at Risk (CVaR).

To begin with, one of the same cost formulation can be revised by bringing in learning-based physical parameter estimation that will empower the system to directly deduce traction or slip through the use of visual and proprioceptive cues instead of relying on prior assigned material properties.

Moreover, one is still to be made when it comes to the extension of the framework by the addition of dynamic and deformable terrains (e.g., loose soil or mud) which would be greatly appreciated by end-users as it would not only make the simulation more realistic but the application areas would also widen. Such an extension would entail the provision of real-time cost-map updates in the light of the feedback from vehicle–terrain interaction.

It is significant to point out that the proposed approach is validated through high-fidelity simulation, however the reality of deploying and benchmarking in the real world is the main next step, which includes transfer learning to reduce the simulation-to-reality gap and evaluation on physical robotic platforms. Yet, these extensions would make the framework take another step forward towards the practical, long-term autonomy in complicated outdoor environments.

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